

Student: Jonatan Rabjerg
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Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 5

Does the sequence $\{a_n\}$ converge or diverge? Find the limit if the sequence is convergent.

$$a_n = -2 + (-1)^n$$

The sequence $\{a_n\}$ converges to the number L if for every positive number ε there corresponds an integer N such that for all n , $n > N \Rightarrow |a_n - L| < \varepsilon$. If no such number L exists, then $\{a_n\}$ diverges. If $\{a_n\}$ converges to L , write $\lim_{n \rightarrow \infty} a_n = L$, or simply $a_n \rightarrow L$, and call L the limit of the sequence.

Note that the sequence looks like $\{-3, -1, -3, -1, \dots, -2 + (-1)^n, \dots\}$. It appears that this sequence will diverge, so test for divergence.

Choose $\varepsilon = \frac{1}{2}$. If the sequence converges, then all terms a_n for n larger than some N must lie within $\varepsilon = \frac{1}{2}$ of L . For all odd n , a_n is -3 .

Therefore, if the sequence converges, -3 must be within $\varepsilon = \frac{1}{2}$ of L . Write this condition as an inequality in the form $|a_n - L| < \varepsilon$.

$$|(-3) - L| < \frac{1}{2}$$

Solve the inequality for L .

$$-\frac{7}{2} < L < -\frac{5}{2}$$

Likewise, for all even n , a_n is -1 . By the same reasoning, if the sequence converges, -1 must be within $\varepsilon = \frac{1}{2}$ of L . Write this condition as an inequality.

$$|(-1) - L| < \frac{1}{2}$$

Solve the inequality for L .

$$-\frac{3}{2} < L < -\frac{1}{2}$$

The number L cannot lie in both intervals, $\left(-\frac{7}{2}, -\frac{5}{2}\right)$ and $\left(-\frac{3}{2}, -\frac{1}{2}\right)$, because they have no overlap. Therefore, no such limit L exists and the sequence diverges.